

Module 4: Assignment

System architecture, security perimeter, and vulnerability mapping

MCY 630 Security Architecture

Assignment Submission Instructions

Submission requirements: assignment

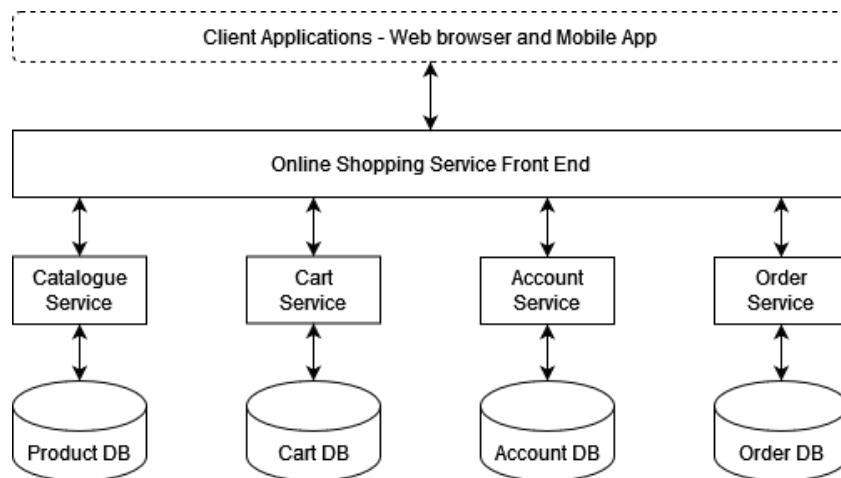
File format: MSWord or pdf (preferred)

Report requirements

- Include your name and assignment title at the top of the document.
- Number the questions and answers and answer each question in your submission in sequence.
- For each response, include the required text as well as any necessary images and diagrams.
- Use black font color for texts
- If uploading an MSWord format file, view the uploaded file from Canvas to make sure that the layout of images and texts are not overlapping in the report. If the contents seem to have moved around, upload a pdf instead of an MSWord file.

Scenario

Consider the following system architecture scenario for a simple online shopping service application



Questions

1. (10 points) Identify and justify with appropriate reasoning on the system architecture design from the application component and service delivery perspectives. (hint: refer to the communication and layout of the components)
2. (10 points) With respect to each of the components in the above system architecture, identify and justify with appropriate reasoning the possible security trust boundaries. (hint: refer to your answer in Q1)
3. (10 points) With respect to the identified security trust boundaries, identify and justify with appropriate reasoning the possible attack surfaces for the given application. (hint: refer to your answer in Q2 and the possible entry/exit points)
4. (20 points) Perform a vulnerability assessment on the identified attack surfaces and the recommended mitigation techniques for the mapped vulnerabilities. (hint: refer to the widely used open-source resources for vulnerability mapping taxonomies)